

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Ans.

1. **Motherboard:** It is the main board of the computer. All other components are connected to it.
2. **CPU (Processor):** It is the main processing unit of the computer. It performs calculations and runs programs.
3. **RAM:** RAM is temporary memory. It helps the computer to run programs faster.
4. **HDD/SSD:** It is used to store files, software, photos, videos and the operating system.
5. **Power Supply (PSU):** It supplies power to all the components inside the computer.
6. **Graphics Card (GPU):** It is used to process images, videos and graphics, especially for gaming.
7. **CPU Cooler:** It keeps the processor cool and prevents it from getting too hot.
8. **Cooling Fans:** Fans help to remove hot air from the cabinet and keep the PC cool.
9. **Optical Drive:** It is used to read CDs and DVDs, if available in the computer.
10. **Expansion Cards:** These cards are added to provide extra features such as Wi-Fi, sound or networking.

2. Disassemble a desktop PC and document each step with clear photos or labeled diagrams and short captions.

Ans.

Step 1: Unplug the Power

First, I switched off the computer and removed the power cable from the PC.



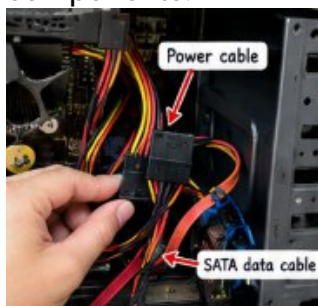
Step 2: Remove the Side Panel

I removed the screws and opened the side panel of the cabinet.



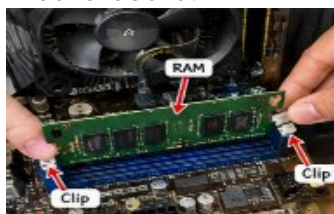
Step 3: Disconnect the Cables

I carefully disconnected the power and data cables connected to different components.



Step 4: Remove RAM

I opened the clips on the RAM slot and carefully removed the RAM from the motherboard.



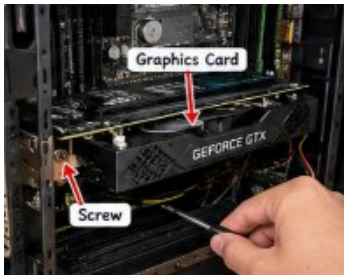
Step 5: Remove HDD/SSD

I disconnected the cables and removed the HDD/SSD from its slot or drive bay.



Step 6: Remove the Graphics Card

I removed the screw and carefully took out the graphics card from the PCIe slot.



Step 7: Remove the Motherboard

I disconnected the remaining connections, removed the motherboard screws, and carefully took out the motherboard.



Step 8: Keep All Parts Safely

Finally, I kept all the removed components and screws safely for reassembly.



3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Ans.

Assemble and Disassemble a PC Practical

Step 1: Install the Motherboard

I placed the motherboard inside the cabinet and tightened the screws properly.



Step 2: Install RAM

I inserted the RAM into the correct slot and pressed it until the clips locked.



Step 3: Install HDD/SSD

I placed the storage drive in its position and connected the power and data cables.



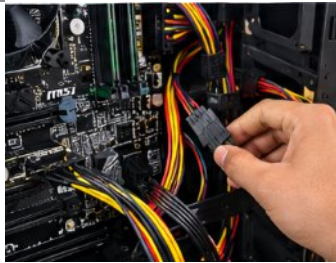
Step 4: Install the Graphics Card

I inserted the graphics card into the PCIe slot and fixed it with a screw.



Step 5: Connect the Cables

I connected the motherboard, storage, graphics card and other required power cables.



Step 6: Connect the CPU Cooler

I connected the CPU cooler properly so that the processor can stay cool.



Step 7: Close the Cabinet

After checking all connections, I placed the side panel back and tightened the screws.



Step 8: Turn On and Check the PC

Finally, I connected the power cable and switched on the PC to check that all components were working properly.



4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Ans.

1. **Switch off the PC and remove the power cable** – This is important to avoid electric shock.
2. **Use a screwdriver** – It is needed to open the cabinet and remove the screws.
3. **Use an anti-static wrist strap** – It helps protect the computer parts from static electricity.
4. **Keep the working table clean and dry** – It helps keep the PC parts safe while working.
5. **Keep all screws in one place** – This helps to avoid losing any screws.
6. **Handle the parts carefully** – Parts like RAM, SSD and motherboard can get damaged if handled roughly.
7. **Do not touch the connector pins** – Touching them can make them dirty or cause connection problems.
8. **Keep water and other liquids away** – Water can damage the PC and cause a short circuit.
9. **Check all connections before starting the PC** – This makes sure everything is connected properly.
10. **Keep the required tools nearby** – It saves time and makes the work easier.